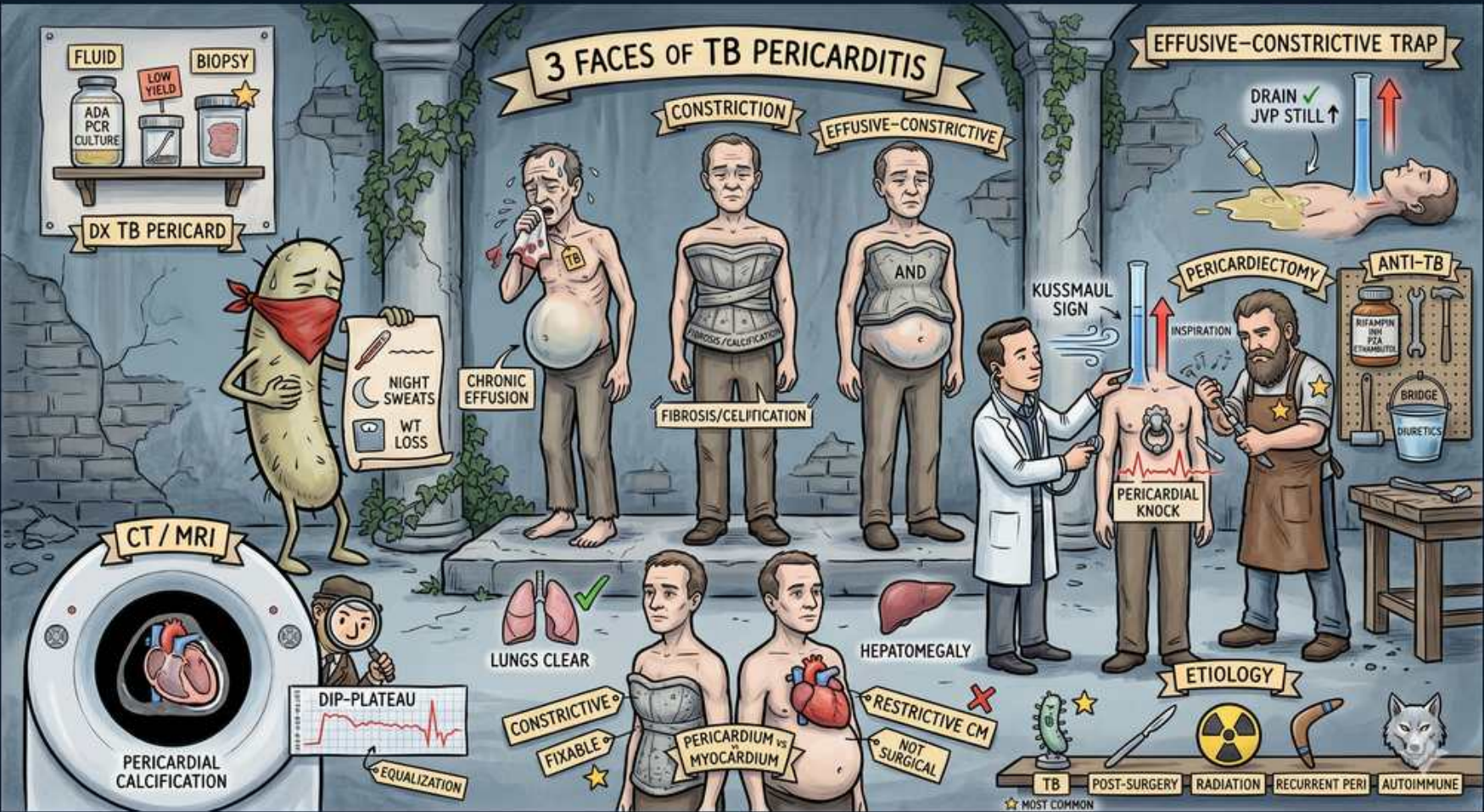


Pericardial Disease — Constrictive Pericarditis



1. 3 faces of TB pericarditis

WHAT YOU SEE

Banner '3 FACES OF TB PERICARDITIS'

WHAT IT ENCODES

Tuberculous pericardial disease presents in three forms: (1) chronic pericardial EFFUSION, (2) CONSTRICTION from a fibrocalcific scar, and (3) EFFUSIVE-CONSTRICTIVE disease combining both. TB is the leading cause of constrictive pericarditis worldwide (especially in endemic and HIV-coinfected populations). Mnemonic: TB pericardium has 3 faces — effuse, constrict, or both.

BOARD TRAP

WRONG to assume a pericardial effusion that drains will resolve all hemodynamic compromise in a TB patient. Discriminator: in effusive-constrictive disease the visceral pericardium is also constricting, so elevated pressures PERSIST after fluid removal — the effusion is only one of the three TB faces.

2. Constriction

WHAT YOU SEE

Middle man wrapped in a tight constricting band 'CONSTRICTION'

WHAT IT ENCODES

Constrictive pericarditis is caused by a rigid, thickened/fibrosed (often calcified) pericardium that encases the heart and impairs DIASTOLIC filling. Early diastole fills rapidly until the ventricle abruptly hits the stiff shell, halting further filling. The result is elevated and equalized diastolic pressures and predominantly right-sided systemic congestion. Mnemonic: a heart in a rigid shell can't expand.

BOARD TRAP

WRONG to confuse constrictive pericarditis with cardiac tamponade. Discriminator: tamponade is an ACUTE fluid problem with pulsus paradoxus and a normal-sized pericardium, whereas constriction is a CHRONIC rigid-pericardium problem featuring Kussmaul sign, pericardial knock, and often calcification — and tamponade lacks the pericardial knock.

3. Fibrosis / calcification

WHAT YOU SEE

'FIBROSIS / CALCIFICATION' label on torso

WHAT IT ENCODES

Chronic inflammation leads to dense fibrosis and dystrophic calcification of the pericardium, converting the normally compliant sac into a stiff, non-yielding case. This stiffening is what mechanically limits ventricular filling and produces the characteristic hemodynamics. Calcification is a strong, fairly specific imaging clue, classically following prior TB.

BOARD TRAP

WRONG to require visible calcification to diagnose constriction. Discriminator: calcification is highly suggestive but present in only a minority; many constriction cases (e.g., post-surgical or post-radiation) show fibrosis WITHOUT calcium, so its absence does not exclude the diagnosis — rely on hemodynamics and imaging of pericardial thickening.

4. Pericardial calcification

WHAT YOU SEE

CT/MRI ring showing 'PERICARDIAL CALCIFICATION'

WHAT IT ENCODES

Pericardial calcification or thickening (>3-4 mm) on CT/MRI — or an 'eggshell' ring on chest X-ray — supports constriction; CT/MRI also assesses pericardial thickness and adherence and helps plan surgery. Classic association is prior TB. Echo adds septal bounce and respiratory variation in mitral inflow. Mnemonic: eggshell heart on imaging = think old TB constriction.

BOARD TRAP

WRONG to use a normal-thickness pericardium on imaging to rule out constriction. Discriminator: up to ~20% of surgically proven constriction has NORMAL pericardial thickness, so a thin pericardium does not exclude it — constriction is ultimately a hemodynamic/physiologic diagnosis confirmed by catheterization and echo Doppler, not by thickness alone.

5. Kussmaul sign

WHAT YOU SEE

Red up-arrow 'KUSSMAUL SIGN' with inspiration

WHAT IT ENCODES

Kussmaul sign = a paradoxical FAILURE of the JVP to fall (or an actual rise) during inspiration. In constriction the rigid pericardium prevents the inspiratory increase in venous return from being accommodated by the right heart, so jugular pressure climbs. It is a classic sign of constrictive pericarditis (also seen in restrictive CM and RV infarction). Mnemonic: Kussmaul — neck veins rise when you breathe in.

BOARD TRAP

WRONG to attribute Kussmaul sign to cardiac tamponade. Discriminator: Kussmaul sign is characteristic of CONSTRICTION (and restriction/RV infarct) but is typically ABSENT in tamponade; tamponade's hallmark is pulsus paradoxus instead. Confusing the two is a classic board trap.

6. Pericardial knock

WHAT YOU SEE

'PERICARDIAL KNOCK' label near auscultated chest

WHAT IT ENCODES

The pericardial knock is an early-diastolic, high-pitched sound (occurring slightly earlier and higher-pitched than an S3) produced by the abrupt halt of rapid ventricular filling against the rigid pericardium. Its presence is fairly specific for constriction. Mnemonic: filling slams into the wall = an audible knock.

BOARD TRAP

WRONG to dismiss the pericardial knock as a benign S3 gallop. Discriminator: the knock is EARLIER and higher-pitched than an S3 and, with elevated JVP and clear lungs, points to constriction rather than the volume-overloaded systolic heart failure that produces an S3.

7. Dip-plateau / square root

WHAT YOU SEE

'DIP-PLATEAU' tracing with equalization

WHAT IT ENCODES

On invasive hemodynamics, constriction shows a 'dip-and-plateau' (square-root sign) in the ventricular pressure tracing — a sharp early-diastolic dip followed by a plateau as filling abruptly stops — and EQUALIZATION of end-diastolic pressures across all four chambers (within ~5 mmHg). Ventricular interdependence with discordant LV/RV systolic pressures during respiration is the most specific finding. Mnemonic: rapid dip, then flat plateau = the rigid wall.

BOARD TRAP

WRONG to use diastolic-pressure equalization to conclude constriction over restrictive cardiomyopathy. Discriminator: restriction can also show a dip-plateau, but in CONSTRICTION the LV and RV systolic pressures move DISCORDANTLY with respiration (enhanced ventricular interdependence), whereas in restriction they move concordantly — this is the key distinguishing finding.

8. Lungs clear

WHAT YOU SEE

Lungs with green check 'LUNGS CLEAR'

WHAT IT ENCODES

In constriction the lungs are typically CLEAR while systemic (right-sided) congestion dominates — elevated JVP, hepatomegaly, ascites, and peripheral edema. This 'clear lungs with high JVP and ascites' pattern is a useful bedside clue that points to a pericardial/right-heart filling problem rather than left-heart failure. Mnemonic: clear lungs but full belly and neck veins = constriction.

BOARD TRAP

WRONG to interpret the picture as left-sided systolic heart failure and diurese as for pulmonary edema. Discriminator: clear lung fields with marked JVP elevation, ascites, and edema indicate a right-sided/pericardial filling problem (constriction), not the pulmonary congestion of left ventricular failure — and the definitive fix is surgical, not diuretic.

9. Hepatomegaly

WHAT YOU SEE

'HEPATOMEGALY' label

WHAT IT ENCODES

Chronic systemic venous congestion from impaired right-heart filling produces hepatomegaly (often pulsatile/congestive), ascites, peripheral edema, and elevated JVP — frequently mistaken for primary liver disease or isolated cirrhosis. Recognizing the elevated JVP alongside the abdominal findings redirects the diagnosis to the pericardium/right heart. Mnemonic: congested liver + high JVP = the heart, not the liver.

BOARD TRAP

WRONG to work up the hepatomegaly and ascites as primary cirrhosis and miss constriction. Discriminator: a markedly ELEVATED JVP accompanying hepatomegaly/ascites indicates cardiac (constrictive) congestion, not intrinsic liver disease — primary cirrhosis does not raise the JVP.

10. Constrictive = fixable

WHAT YOU SEE

'CONSTRICTIVE FIXABLE' vs 'RESTRICTIVE CM NOT FIXABLE'

WHAT IT ENCODES

Distinguishing constrictive pericarditis from restrictive cardiomyopathy is high-yield because constriction is surgically CURABLE by pericardiectomy, whereas restrictive cardiomyopathy (e.g., amyloid) is not. Clues favoring constriction: respiratory variation in mitral inflow (>25%), septal bounce, normal/elevated mitral annular e' (annulus paradoxus), pericardial thickening/calcification, and discordant ventricular pressures. BNP tends to be lower in constriction than in restriction. Mnemonic: constriction = curable shell; restriction = sick muscle.

BOARD TRAP

WRONG to manage a patient with restrictive-cardiomyopathy physiology as if pericardiectomy will help. Discriminator: in restriction the myocardium itself is diseased (e' is REDUCED, BNP markedly elevated, no respiratory inflow variation), so surgery does not help — only true constriction (preserved/elevated e', respiratory variation, pericardial pathology) benefits from pericardiectomy.

11. Pericardiectomy

WHAT YOU SEE

'PERICARDIECTOMY' label

WHAT IT ENCODES

Pericardiectomy (surgical removal of the rigid pericardium) is the only DEFINITIVE treatment for chronic constrictive pericarditis and is offered to symptomatic patients. Operative mortality is significant (especially in radiation-induced or advanced disease), so timing matters — operate before end-stage myocardial/hepatic dysfunction. Diuretics give only temporary symptom relief while awaiting surgery. Mnemonic: peel the shell to free the heart.

BOARD TRAP

WRONG to rely on lifelong diuretics/medical therapy as definitive treatment for established chronic constriction. Discriminator: medical therapy only palliates congestion; pericardiectomy is the sole definitive cure. (Exception: a transient/subacute constriction from active inflammation may resolve with a 2-3 month anti-inflammatory trial before committing to surgery.)

12. Effusive-constrictive

WHAT YOU SEE

Right man 'EFFUSIVE-CONSTRICTIVE AND'

WHAT IT ENCODES

Effusive-constrictive pericarditis combines a pericardial EFFUSION with constriction of the visceral pericardium. The tell-tale feature: after pericardiocentesis removes the fluid and intrapericardial pressure normalizes, the right atrial pressure remains ELEVATED because the visceral layer is still constricting. Management may require visceral pericardiectomy. Mnemonic: drain the fluid but the pressure stays high.

BOARD TRAP

WRONG to expect pericardiocentesis to fully resolve hemodynamics in effusive-constrictive disease. Discriminator: persistently elevated right atrial pressure DESPITE normalized intrapericardial pressure after drainage defines effusive-constrictive physiology — the constricting visceral layer remains and may need surgery.

13. Anti-TB therapy

WHAT YOU SEE

'ANTI-TB' shelf on right

WHAT IT ENCODES

Tuberculous pericarditis is treated with standard multidrug anti-tuberculous therapy (RIPE: rifampin, isoniazid, pyrazinamide, ethambutol). Adjunctive corticosteroids may reduce constriction/reaccumulation in selected HIV-negative patients (benefit is debated). If constriction develops or persists despite adequate anti-TB therapy, pericardiectomy is indicated. Mnemonic: treat the bug (RIPE), then cut the shell if it still constricts.

BOARD TRAP

WRONG to assume anti-TB drugs alone will reverse established fibrocalcific constriction. Discriminator: medical anti-TB therapy treats the active infection and may prevent progression, but once a rigid constrictive scar has formed, only pericardiectomy relieves it — antibiotics will not peel a calcified pericardium.

14. Etiology bar

WHAT YOU SEE

'ETIOLOGY': TB, post-surgery, radiation, recurrent peri, autoimmune

WHAT IT ENCODES

Causes of constrictive pericarditis: TUBERCULOSIS (the leading cause worldwide / in endemic regions), and in developed countries idiopathic/viral, prior CARDIAC SURGERY, mediastinal/chest RADIATION (e.g., for lymphoma or breast cancer), recurrent pericarditis, connective-tissue/autoimmune disease, and uremia. Mnemonic: TB and Radiation and Surgery and Idiopathic = the big four.

BOARD TRAP

WRONG to think only of TB and overlook the iatrogenic causes that dominate developed-world practice. Discriminator: prior cardiac surgery and mediastinal radiation are leading causes in developed countries and notably radiation-induced constriction often coexists with myocardial damage, carrying higher pericardiectomy risk and worse outcomes than TB-related constriction.